
Simulation of Partial Melting Mantle

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Mantle melting:

1. Mantle melting is a crucial geophysical process in lithospheric evolution and plate tectonics.
2. Geophysical evidence indicates that the asthenosphere is often partially molten.
3. A thorough understanding of partial melting is fundamental to comprehending the dynamics of magmatism.

Partial Melting:

1. Modeled as a two-phase flow framework.
2. Coupled with complicated chemical interactions.
3. Compaction is essential in deciding melt extraction and segregation.

Challenge:

1. Degeneracy caused by no-melt regions poses mathematical and numerical difficulties.

Model

1. A degenerate two-phase Darcy–Stokes system describing the coupled interaction between the solid matrix and the interstitial melt.
2. A nonlinear system governing the transport of chemical species and thermal energy.
3. A eutectic phase relation characterizing the melting processes and connecting the transport and static mechanical system through enthalpy method.

Numerics

1. $H(\text{div})$ - and H^1 -conforming mixed finite element spaces for Darcy and Stokes subproblems.
2. A more versatile and flexible multilevel WENO weighting scheme to address the nonlinear transport of chemical species and energy in the presence of porous media.

Darcy-Stokes System

Full static system:

$$\phi_l(\mathbf{v}_l - \mathbf{v}_s) + \frac{k\phi_l}{\mu_l}(\nabla p_l - \rho_f \mathbf{g}) = 0, \quad (1)$$

$$\nabla \cdot (\phi_l \mathbf{v}_l + \phi_s \mathbf{v}_s) = 0, \quad (2)$$

$$\nabla \bar{p} - \nabla \cdot \boldsymbol{\tau}(\mathbf{v}_s) + \bar{\rho} \mathbf{g} = 0, \quad (3)$$

$$\mu_s \nabla \cdot \mathbf{v}_s - \phi_l(p_l - p_s) = 0, \quad (4)$$

Deviatoric stress:

$$\boldsymbol{\tau}(\mathbf{v}_s) = 2\mu_s \phi_s \left(\mathcal{D} \mathbf{v}_s - \frac{1}{3} \nabla \cdot \mathbf{v}_s \mathbf{I} \right), \quad (5)$$

